

Small Models, Big Reactions: Exploring the Performance–Cost Tradeoff in Emoji Prediction

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1 Problem and Motivation

Emoji are widely used in online communication to express emotion, tone, humor, and reactions that may not be obvious from text alone. We study emoji prediction as a multi-class text classification task: given a short social-media post with its emoji removed, the model predicts one of a fixed set of emoji classes.

Prior work has shown that both simple classifiers and pretrained language models can perform this task, but these approaches vary greatly in predictive performance and computational cost. Understanding this tradeoff is useful because a smaller model may be preferable when computational resources or inference speed are limited, even if it sacrifices some predictive performance. We therefore study two questions:

RQ1: What is the tradeoff between emoji-prediction performance and model complexity, measured by macro-F1, parameter count, and inference latency?

RQ2: Are the performance gains of more complex models concentrated in difficult or semantically similar emoji classes?

2 Literature Survey

SemEval-2018 Task 2 formalized emoji prediction as predicting the most likely emoji associated with a tweet and used macro-F1 to account for class imbalance [2]. Submitted systems ranged from traditional classifiers to convolutional and recurrent neural networks, showing that substantially different model families can be applied to the same prediction task.

TweetEval later incorporated emoji prediction into a standardized Twitter classification benchmark with fixed data splits and pretrained language-model baselines [1]. Its results showed the effectiveness of pretrained Transformer representations for social-media classification.

Our model choices span several levels of complexity. TF-IDF with logistic regression provides a lightweight baseline, while BiLSTM introduces recurrent sequence modeling [4, 6]. BERT provides a large pretrained Transformer baseline [3], while DistilBERT was designed as a smaller and faster version of BERT [5]. These models therefore provide a natural spectrum for studying the performance–cost tradeoff.

3 Data Requirements

We will use the TweetEval emoji classification benchmark [1], which contains short social-media posts labeled with one of 20 emojis. The provided splits contain approximately 45,000 training, 5,000 validation, and 50,000 test examples. The dataset directly matches our task and enables comparison with established baselines.

TweetEval contains substantial class imbalance and semantically similar emoji classes, making macro-F1 and class-level error analysis important. We will preserve the provided splits and apply only limited preprocessing, such as removing residual Unicode artifacts when necessary.

4 Experiment Plan

We will compare four models of increasing complexity: TF-IDF with logistic regression as a lightweight baseline, a BiLSTM classifier, DistilBERT, and BERT-base. All models will use the same TweetEval train, validation, and test splits. Neural models will be tuned using a small hyperparameter search on the validation set, and we will run multiple random seeds to reduce sensitivity to initialization.

Following TweetEval, macro-F1 will be our primary evaluation metric, with accuracy reported as a secondary measure. Computational cost will be measured using parameter count and inference latency. Latency will be measured on the same hardware with a fixed batch size after warm-up runs and averaged across repeated measurements. We will also analyze confusion matrices and representative errors to determine whether more complex models better distinguish semantically similar emoji classes.

5 Feasibility and Next Steps

We will run the experiments primarily in Google Colab Pro. An NVIDIA T4 GPU will be our expected training resource, while an A100 may be used when available to reduce training time. Our implementation will rely on Python, scikit-learn for the TF-IDF baseline, PyTorch for neural models, and Hugging Face Transformers and Datasets for pretrained models and data loading.

Our first experiment will establish the TF-IDF + logistic regression baseline and verify the complete TweetEval training and evaluation pipeline. By the milestone, we plan to evaluate the baseline, BiLSTM, and at least one pretrained Transformer and report preliminary macro-F1, parameter-count, and inference-latency results. By the final report, we will complete the full comparison across all planned models and perform class-level and qualitative error analysis.

The main dependencies are access to the TweetEval data, the required Python libraries, and Colab GPU availability. Potential obstacles include class imbalance, noisy social-media text, preprocessing artifacts, and variation in GPU availability. If compute becomes limiting, we will reduce the number of larger Transformer experiments while preserving the core performance-complexity comparison.

References

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